

A. Multiple Choice [60%, each 3%]

1. D

Which of these truth tables represents the Exclusive-NOR gate?

A	B	X	A	B	X	A	B	X	A	B	X
0	0	1	0	0	1	0	0	0	0	0	1
0	1	0	0	1	1	0	1	1	0	1	0
1	0	0	1	0	1	1	0	1	1	0	0
1	1	1	1	1	0	1	1	0	1	1	0

(A)

(B)

(C)

(D)

A) (A)

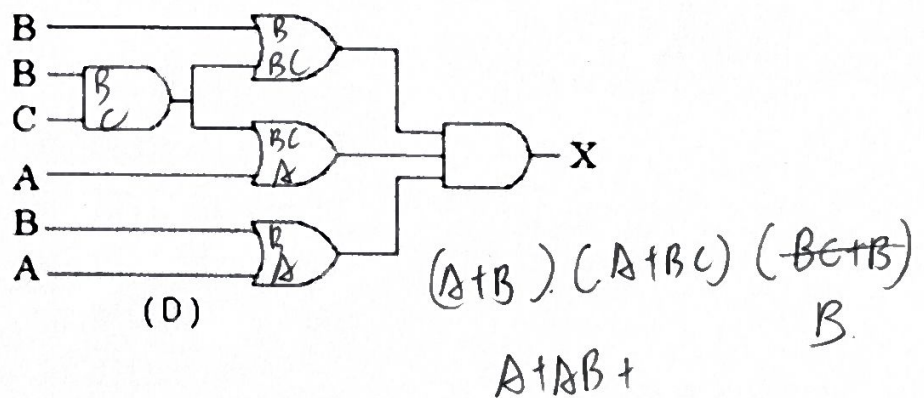
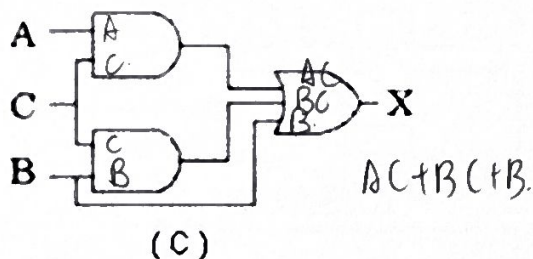
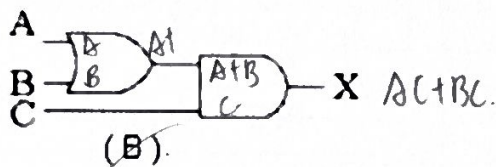
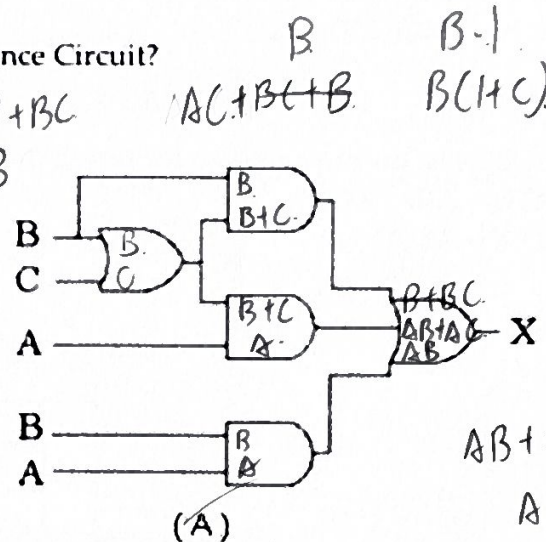
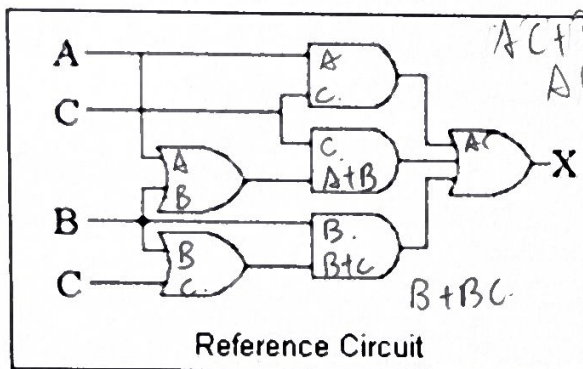
B) (B)

C) (C)

D) (D)

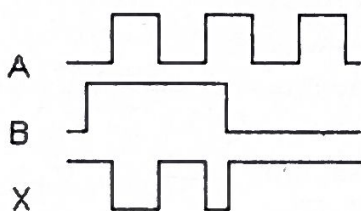
2. C 11/15/2014

Which of the circuits below is equivalent to the Reference Circuit?



3. B

This is the timing diagram for a 2-input _____ gate.



A) Exclusive-OR

B) NAND

C) OR

D) AND

4. C

Which type of gate can be used to add two bits?

A) NAND

B) XNAND

C) XOR

D) NOR

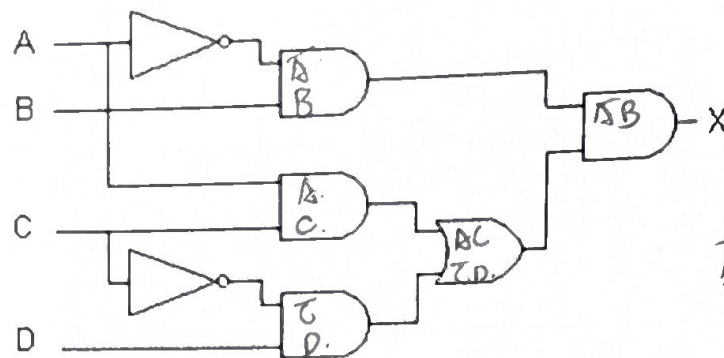
5. B

Figure 5-2

A correct logic expression for Figure 5-2 is _____.

A) $X = (\overline{A}B)(AC + \overline{C}D)$

B) $X = (\overline{A}B)(A\overline{C}CD)$

C) $X = ABC(\overline{C}BD)$

D) $X = \overline{A}BC + A\overline{C}D$

6. C

The simplest output equation for Figure 5-2 is _____.

A) $X = \overline{A}(\overline{B}C + CD)$

B) $X = \overline{A}BC + B\overline{C}D$

C) $X = \overline{A}BCD$

D) $X = \overline{A}(BCD)$

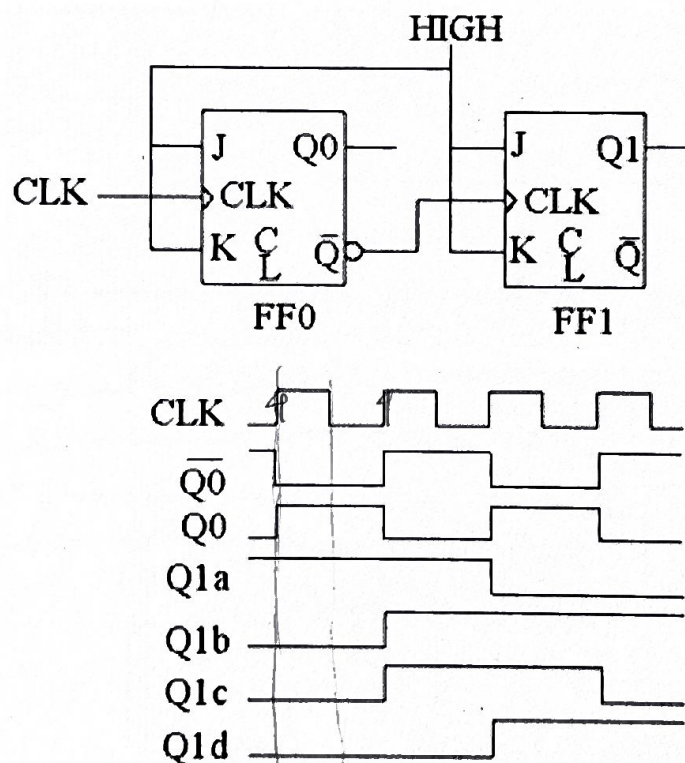
7. C

Figure 8-1

The counter shown in Figure 8-1 is a(n) _____ counter, and the correct output waveform for Q1 is _____.

A) ~~synchronous~~, Q1a

B) asynchronous, Q1b

C) asynchronous, Q1c

D) ~~synchronous~~, Q1d

8. A

Which of the following is a form of DeMorgan's theorem?

A) $\overline{XY} = \overline{X} + \overline{Y}$

B) $X + 0 = 0$

C) $\overline{X + Y} = \overline{X} + \overline{Y}$

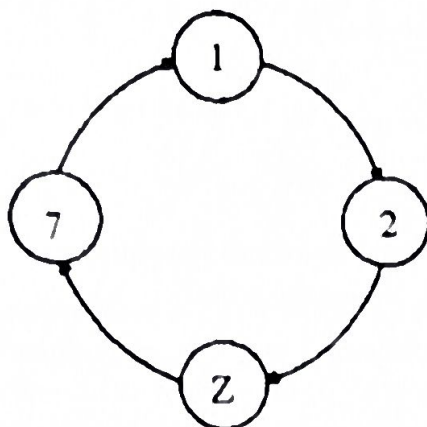
D) $X(1) = X$

9. D

Which of the following expressions is in the sum-of-products form?

- A) $(AB)(CD)$ B) $(A + B)(C + D)$ C) $AB(CD)$ D) $AB + CD$

10. B



Present State			Next State		
Q2	Q1	Q0	Q2	Q1	Q0
0	0	1	0	1	0
0	1	0	1	0	1
1	0	1	1	1	X
1	1	1	0	0	1

Figure 8-8

Refer to Figure 8-8. What value should 'X' represent in the state table?

- A) 0 B) 1

What value does 'Z' represent in Figure 8-8?

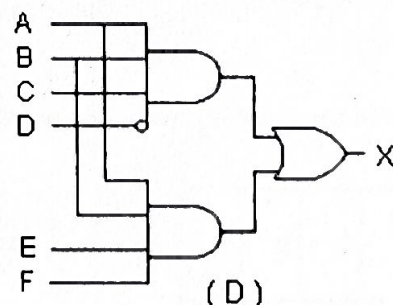
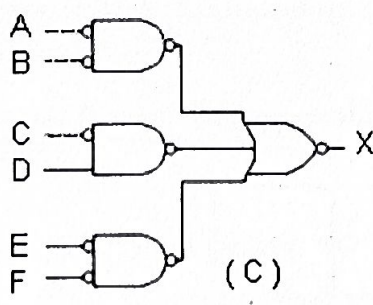
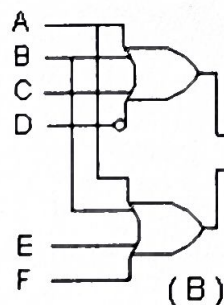
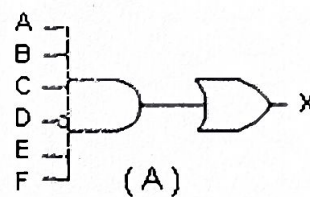
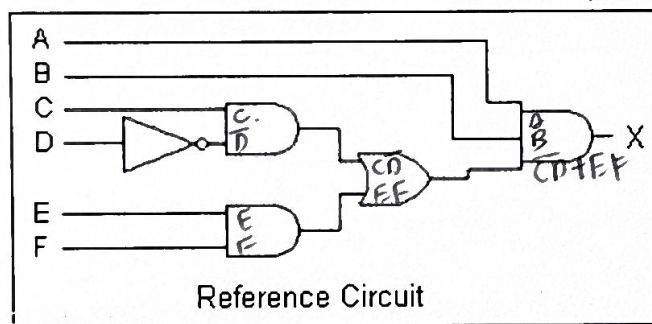
11. B A) 111 B) 5 C) 3 D) 4

$$2^2 + 2^0 = 5$$

12. D

Which circuit is the sum-of-products equivalent of the Reference Circuit?

$$AB(C\bar{D}EF) = AB\bar{C}\bar{D}EF + ABCE\bar{F}$$



- A) Figure(A) B) Figure (B) C) Figure (C) D) Figure (D)

13. C

NOR

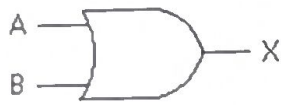
The relationship between a NAND gate and a negative-OR gate is expressed by _____.

- A) $AB = \overline{A + B}$ B) $\overline{A + B} = \overline{A} + \overline{B}$ C) $\overline{AB} = \overline{A} + \overline{B}$ D) $AB = \overline{\overline{A} + \overline{B}}$

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14. C

This is the logic symbol for a(n) _____.



- A) Inverter
C) OR gate

- B) AND gate
D) None of the above

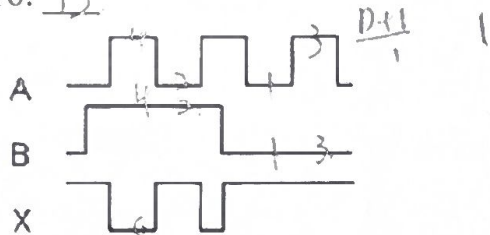
15. C

How many gates, including inverters, are required to implement the equation,
 $X = ACD + AB(CD + BC)$, after simplification with Boolean algebra?

A) 9

B) 7

D) 5

16. B

A) Exclusive-OR

B) NAND

C) OR

D) AND

17. D

How many flip-flops are required to construct a decade counter?

A) 8

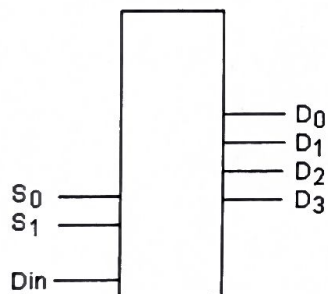
B) 5

C) 10

D) 4

18. C

The circuit below is most likely a _____.

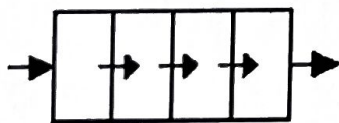


A) full-adder

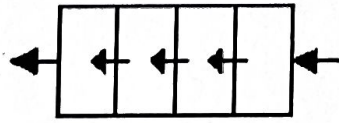
B) comparator

C) demultiplexer

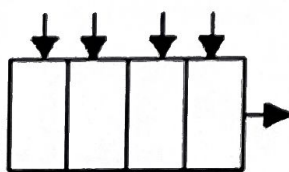
D) multiplexer

19. CWhich of the illustrations below represents the parallel-in serial-out register function?

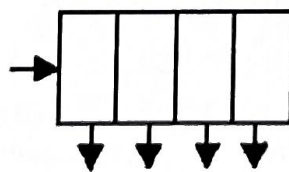
I.



II.



III.



IV.

A) Figure I.

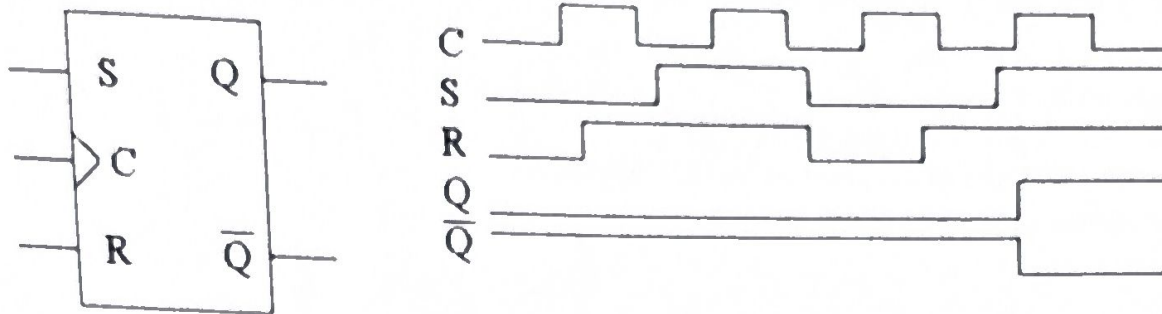
B) Figure II.

C) Figure III.

D) Figure IV.

20. D

Given the waveforms for this R-S flip-flop, what is wrong with the circuit?



- A) The outputs should have toggled on the leading edge of the second clock pulse; the IC is bad.
- B) The outputs should change on the trailing edge of the clock; the clock signal is inverted.
- C) The Q output should be high at the beginning of the second clock pulse. The IC is defective.
- D) When both R and S are high at the same time, the output is unpredictable. Nothing is wrong with the circuit.

B. Short Answer [40%]

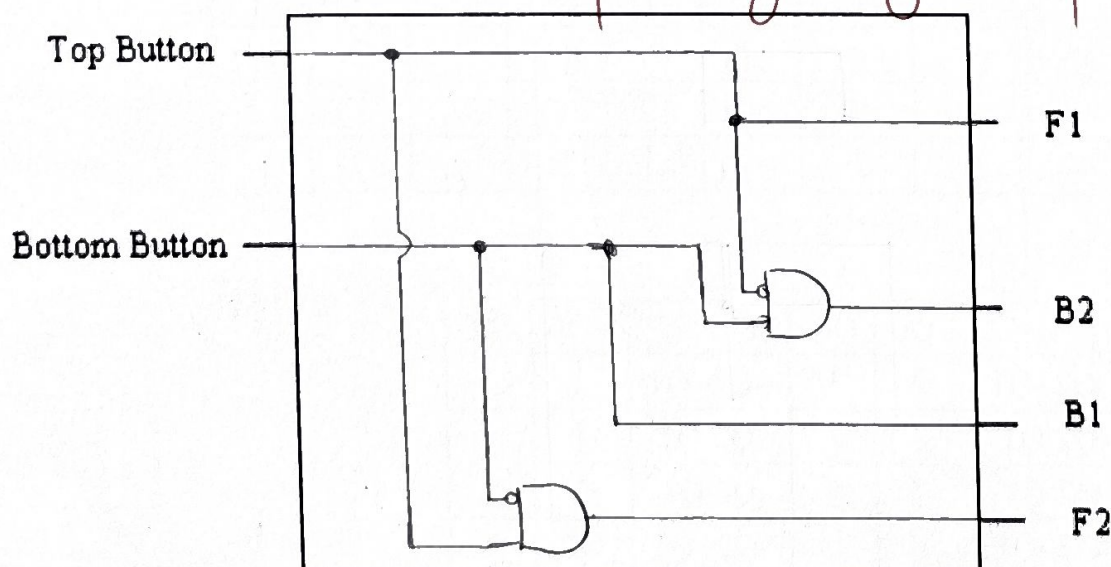
1.

Design the logic to control the motor on a simple remote control car. There are two buttons on the remote control for the motor. If neither button is pressed, the car stops. If the top button is pressed, the car moves forward. If the bottom button is pressed, the car moves backwards. If both buttons are pressed, the car spins. There are two motors in the car and each motor has two inputs F and B. A high on F makes the car go forward. A high on B makes the car go backward.

A: Make a truth table for this design [5%]

State	Top button	Bottom button	F1 for motor 1	B1 for motor 1	F2 for motor 2	B2 for motor 2
Stop	0	0	0	0	0	0
Backward	0	1	0	1	0	1
Forward	1	0	1	0	1	0
Spin	1	1	1	1	0	0

B: From the truth table, draw your design with gate symbols [5%]



$F_1 = \overline{B}T$
 $B_1 = B$
 $F_2 = T$
 $B_2 = \overline{T}B$
 不會同時往前往後
 $F_1 > F_1$
 $F_2 = \overline{T}B$
 $B_2 = \overline{T}B$

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2. [30%]

Design a synchronous counter to produce the following binary sequence. Use J-K flip-flops,

0, 1, 2, 7, 3, 6, 4, 5, 0

a) Draw the state diagram (6 points)

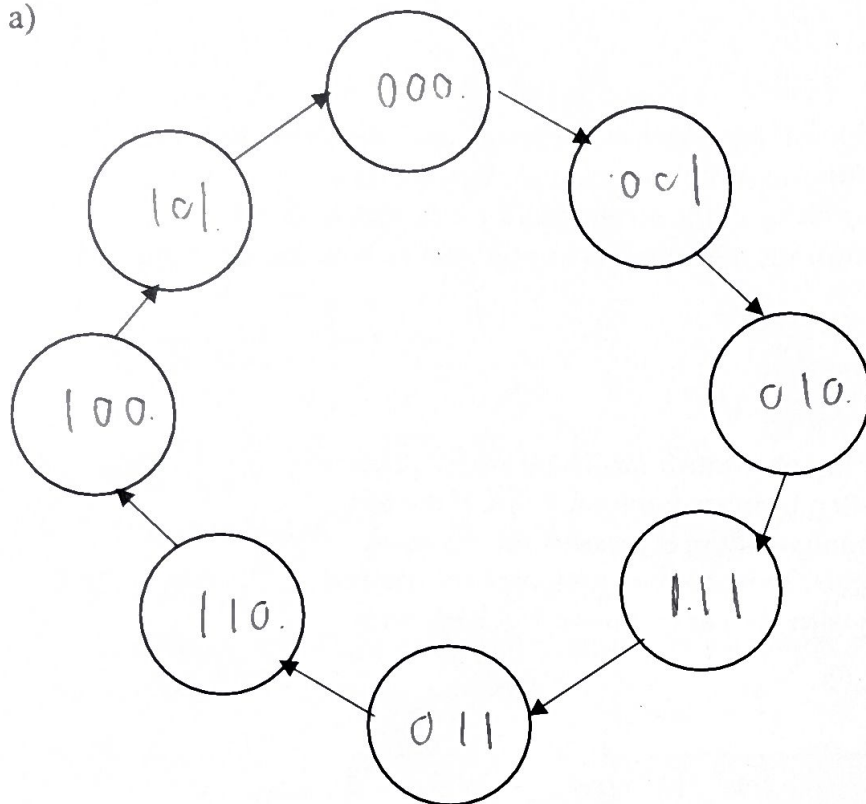
b) List the next-state table (6 points)

c) List the K-maps of all J's and K's (6 points)

d) Write down the Boolean expressions of all J's and K's (6 points)

e) Draw the logic diagram of your design (6 points)

a)



	J	K
0 → 0	0	X
0 → 1	1	X
1 → 0	X	1
1 → 1	X	0

b)

Q2	Q1	Q0	Q2'	Q1'	Q0'	J2	K2	J1	K1	J0	K0
0	0	0	0	0	1	0	0/1	0	0/1	1	0/1
0	0	1	0	1	0	0	0/1	1	0/1	0/1	1
0	1	0	1	1	1	1	0/1	0/1	0	1	0/1
1	1	1	0	1	1	0/1	1	0/1	0	0/1	0
0	1	1	1	1	0	1	0/1	0/1	0	0/1	1
1	1	0	1	0	0	0/1	0	0/1	1	0	0/1
1	0	0	1	0	1	0/1	0	0	0/1	1	0/1
1	0	1	0	0	0	0/1	1	0	0/1	0/1	1

c)

J2	~Q0	Q0	J1	~Q0	Q0	J0	~Q0	Q0
~Q2~Q1	0	0	~Q2~Q1	0	1	~Q2~Q1	1	X
~Q2Q1	1	1	~Q2Q1	X	X	~Q2Q1	1	X
Q2Q1	X	X	Q2Q1	X	X	Q2Q1	0	X
Q2~Q1	X	X	Q2~Q1	0	0	Q2~Q1	1	X

$$J_2 = Q_1$$

$$J_1 = \overline{Q_2}Q_0$$

$$J_0 = \overline{Q_2} + Q_2\overline{Q_1}$$

K2	$\sim Q0$	Q0	K1	$\sim Q0$	Q0	K0	$\sim Q0$	Q0
$\sim Q2 \sim Q1$	X	✓	$\sim Q2 \sim Q1$	X	X	$\sim Q2 \sim Q1$	X	X
$\sim Q2 Q1$	X	X	$\sim Q2 Q1$	0	0	$\sim Q2 Q1$	X	1
Q2 Q1	0	1	Q2 Q1	1	0	Q2 Q1	X	0
Q2 $\sim Q1$	0	1	Q2 $\sim Q1$	X	X	Q2 $\sim Q1$	X	1

d)

$$J2 = Q_1 \cdot J_1$$

$$K_2 = G_0.$$

$$J1 = \overline{Q_2} Q_0$$

$$K1 = G_2 \overline{G_0}$$

$$J0 = \overline{Q_1} + Q_2 \overline{Q_1}$$

$$K0 = \overline{G_2} + \overline{G_1}$$

e)

$$\overline{Q_2} + \overline{Q_1}$$

